

Frīgedæg

GRAINS · TWO REELS · A RING OF FIVE CELLS

Grains over a recording

Frīgedæg runs grains over a recording. There is no synthesiser inside — everything you hear is a fragment of the material. Two reels and a ring of five cells keep the whole thing moving without a finger on it.

A test tone is already loaded at launch, so it makes a sound straight away. Record something with **MIC**, or load a file, and the eight pads read it back as grains — where, how long, how high, how scattered, and through which filter and effect.

Behind the pads sits **RENTHOUSE**: a ring of five petals passing a held voltage around, and two loopers whose bookmarks, direction, speed and level are driven by the ring. What the grains make is what the reels record, and what the reels play is material again.

Too much to read? Skip it. Ignore MIXER… entirely. Hold MIC, say something, let go, and move the pads — that is the whole instrument. The ring and the loopers are there for when you want the sound to keep moving on its own.

What it needs

IOS	iOS 18.0 or later.
DEVICE	iPhone and iPad, played in landscape. On iPad the whole instrument fills the screen and every panel opens full size.
MICROPHONE	Only while MIC or LIVE is on.
CAMERA	Only when CAM is on; the picture is read as movement and light, nothing is kept.
NETWORK	None. Nothing is sent anywhere.

Contents

02 The front — keys and pads

03 The sound — material, playing, engines, LEN / RGN

04 The keys

05 The pads

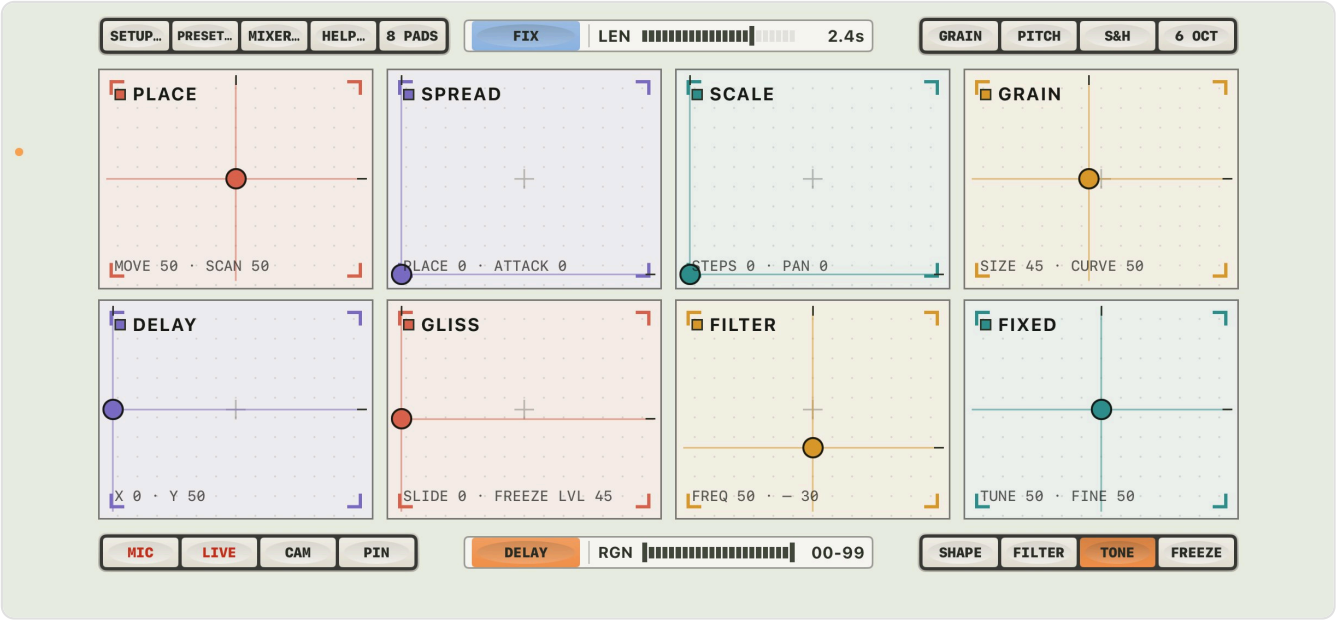
06 SETUP… and PRESET…

07 RENTHOUSE — the ring

08 The loopers

09 The filter at the exit, tips, credits

Keys and pads



Two rows of keys frame eight pads. Each group of keys sits in its own dark frame and has its own colour; a key that is on is filled with that colour. Keys with "…" open a panel. Keys with several names step through them on every press, and the last step is off.

TOP LEFT	**SETUP…** the desk · **PRESET…** camera, materials, presets · **MIXER…** RENTHOUSE, the ring and the loopers · **HELP…** the manual inside the app · **8 PADS / 4 PADS** the layout.
TOP CENTRE	How the grains are played (tap for the next, hold for the one before) and **LEN**, the longest a grain can be.
TOP RIGHT	The engine, and the pitch — **PITCH**, **S&H**, the range (**6 / 8 / 10 OCT**).
BOTTOM LEFT	The ways in — **MIC**, **LIVE**, **CAM**, **PIN**.
BOTTOM CENTRE	The effect (tap for the next, hold for back) and **RGN**, the part of the material that is read.
BOTTOM RIGHT	Shaping the sound — **SHAPE**, **FILTER**, **TONE**, **FREEZE**.

Material, playing, engines

Getting material in

MIC records only while it is held; what you played up to release becomes the new material, normalised to its peak. **LIVE** keeps recording all the time and turns what comes in straight into grains — an effect rather than a sampler. A file is loaded from PRESET..., into slot A or B.

How the grains are played

The left end of the top bar. Tap for the next, hold for the one before.

FIX	Grains laid out in an even stream from the read point. No random numbers are drawn.
SCATTER	Spacing and position scattered — a cloud. Moving a finger fast bows it: more grains, shorter.
TOUCH	Each pad is a surface to strike. Where you touch sets that grain's length, pitch (up to two octaves either way) and place.
LFO	The ring in RENTHOUSE drives the grains.
GATE	A grain is thrown every time the ring's gate opens.
RIPPLE	A touch drops a seed that keeps throwing grains, spreading outward and fading.
LOOP	The two loopers become an 8 second ring, and what comes out is fed back in as the next material.

The engines

The first blue key on the top row. **GRAIN** is plain overlap. **PULSE** is pulsar synthesis — the rate grains are struck at is the pitch and the grain length is the formant, set apart from each other (the GRAIN pad becomes **PULSAR**). **FEED** reads from a six second ring that its own output is written back into.

LEN and RGN

LEN is the ceiling on the length of a grain (the GRAIN pad's X runs up to it). Below 20 ms a grain stops being a sample and becomes a timbre. **RGN** is the part of the material that is read: drag near either end to move that end.

Pitch, shape, filter, tone

Pitch

PITCH	Every grain takes its own pitch from nine clean ratios and glides up or down inside the grain, up to an octave.
S&H	A phrase: one ratio from the same nine (two octaves either way) is held for 4 to 16 grains, then another is drawn. The windows overlap, so the steps join smoothly.
6 / 8 / 10 OCT	The width of the pitch pad, either way.

SHAPE

PERC	Struck: an instant attack, then a steep fall.
SUS	The grain circles a 20 ms fragment and lasts at least 0.6 s — a held tone, strongest up high.
REV	Every grain backwards.

FILTER

Only the grains pass through it. One key steps through the kinds, and the FILTER pad changes its two knobs to suit. It never cuts the sound away entirely — the floor is −22 dB and what is lost is brought back up.

LP / HP	Low pass / high pass. X the frequency, Y the resonance.
BP	Band pass. X the centre, Y how narrow.
COMB	Teeth spaced by X; Y makes them sharper.
RING	Multiplied by a sine, 20 Hz to 10 kHz. Y is the depth.
SHIFT	Every frequency moved by the same amount — left down, right up. Y feeds it back into a spiral.
CRUSH	X holds samples, Y takes bits away.
STUT	Cuts the last moment and repeats it now and then. X how often, Y how long.

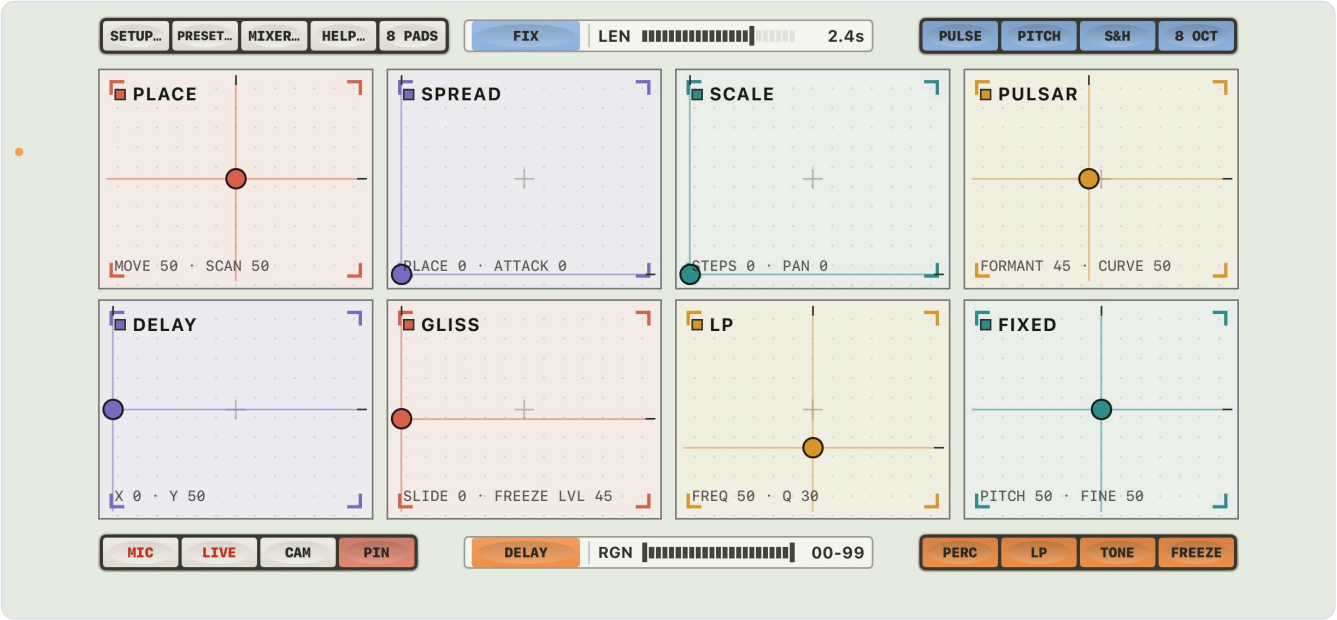
TONE, FREEZE, the effect

TONE is the colour on the way out — a lift above 1 kHz and a soft saturation. **FREEZE** holds the reverb tail and keeps it circling. The effect is chosen at the left end of the bottom bar: DELAY, VERB, HARM, FOLD, DIGI, or NO FX. Its pad is its two knobs.

PIN and CAM

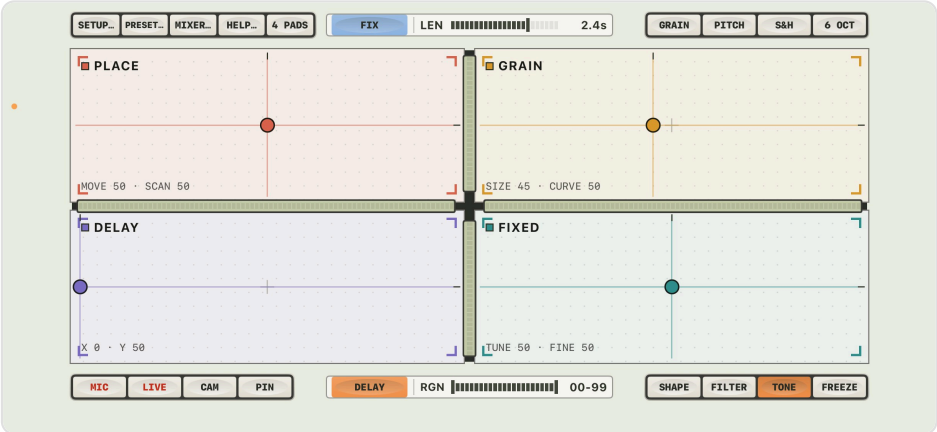
PIN holds the read point where it is: every grain comes from that one place. **CAM** lets the camera move the pads — each pad follows the most moving, brightest or darkest part of its own square of the picture (chosen at the top of PRESET…).

Eight squares, or four



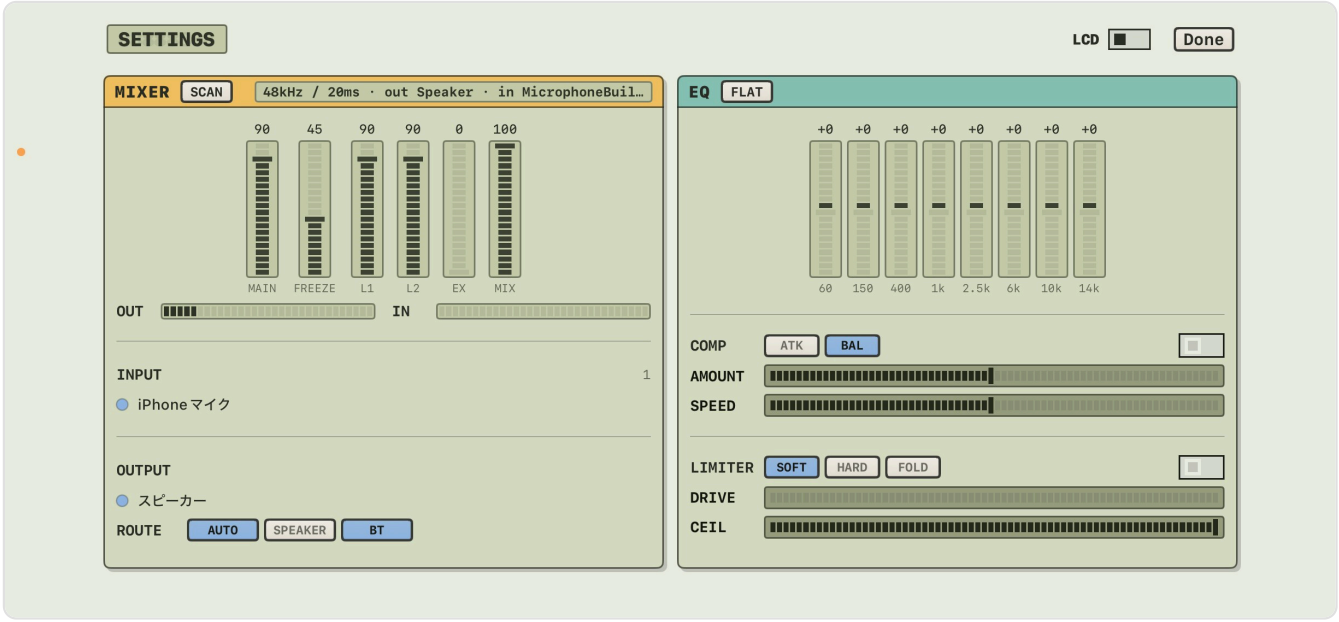
PLACE	X moves the read point, Y how fast it travels on its own (centre stops).
SPREAD	X how far grains scatter from the read point, Y the attack.
SCALE	X the steps: 1 to 8 pitches walked in order (root, octave, fifth …). Y the stereo spread.
GRAIN / PULSAR	X the grain length (the formant in PULSE), Y how the ramps bend.
EFFECT PAD	The chosen effect's two knobs (it is named after the effect — DELAY, VERB …).
GLISS	X how far each grain slides (up to two octaves, up or down per grain). Y the FREEZE level.
FILTER PAD	The two knobs of whatever the FILTER key is on (named after it — LP, BP …).
PITCH PAD	X the pitch (width set by the OCT key), Y a fine trim. Its name follows how the grains are played.

4 PADS

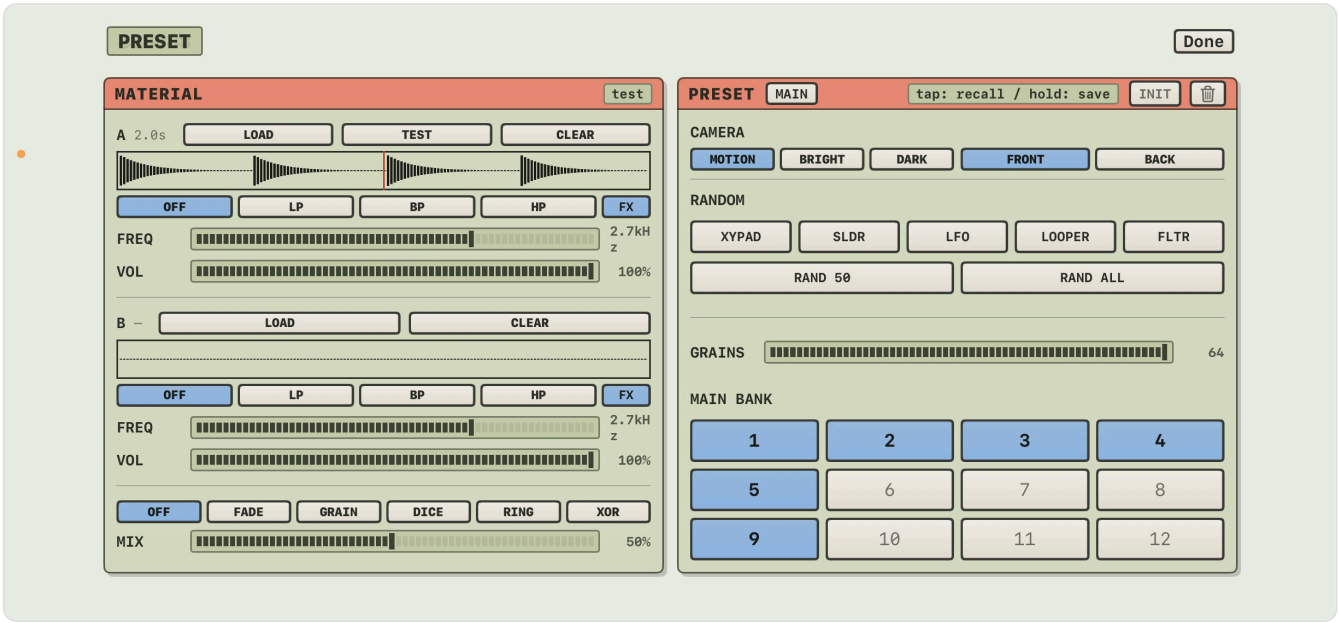


The original layout: PLACE, GRAIN, the effect and the pitch, with the cross sliders between them — left the attack, right the spread, down the scale steps, up the stereo spread. Everything the inner pads do is still there.

SETUP… and PRESET…



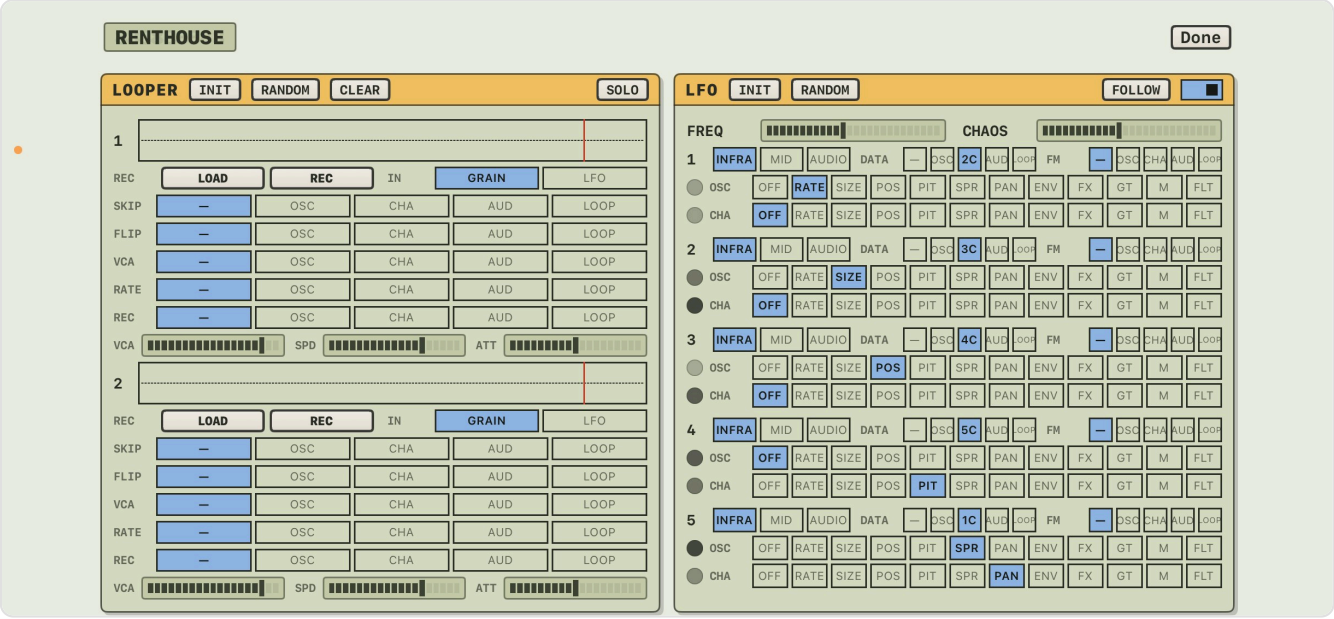
SETUP… is the desk: faders (MAIN, FREEZE, the two loopers L1 / L2, EX, MIX), the OUT / IN meters, the audio input and output and the route (AUTO, SPEAKER, BT), and the EQ, COMP and LIMITER at the exit. The LCD switch in the heading turns the pads' LCD look on or off.



PRESET… — on the left the two materials, A and B: LOAD, TEST (A), CLEAR, a filter for each (OFF, LP, BP, HP, FX) with FREQ and VOL, and how the two are mixed (OFF, FADE, GRAIN, DICE, RING, XOR). On the right the camera (MOTION, BRIGHT, DARK · FRONT, BACK), the keys that shake it, the number of grains and the bank.

SLOTS 1–12	Tap to recall, hold to save; press the bin then a slot to clear it. A preset keeps the whole instrument and survives quitting the app. Hold INIT to put everything back.
XYPAD · SLDR · LFO · LOOPER · FLTR	Each redraws one corner of the instrument. RAND 50 picks two or three corners at random; RAND ALL redraws the lot.
HELD, NOT TAPPED	The patching and modes jump at once; everything that moves continuously travels there over about five seconds. Let go early and it arrives at once.

The ring of five



Opened with **MIXER…**. Five petals stand in a ring. Each has a triangle oscillator and two follow-and-hold circuits struck by that triangle's square: while the square is high the circuit chases its neighbour, while it is low it grabs and stops. There is no randomness anywhere in it. The switch on the LFO card unplugs the whole ring at once — the patching stays where it is.

INPUTS

DATA	Every rising edge steps who that petal is chasing on by one — the waveform folds and the pulse skips.
FM	Moves the speed directly, always continuous.
AUD	The instrument's own output; press again for EX, what comes back from the exit filter.
RANGE	INFRA, MID, AUDIO. Up at AUDIO it becomes pitch.

OUTPUTS

OSC is the triangle, CHA the battlement. Choose a destination per petal — grain parameters (RATE, SIZE, POS, PIT, SPR, PAN, ENV), the effect (FX), the gate (GT), the ring itself (M) or the exit filter (FLT). Press the destination you are on to step inside its group. Nothing happens until a destination is chosen.

RESET	Puts the ring back to how it started; a watchdog also returns it after four seconds.

Two reels, always turning

The reels reach the end, go back to the top, and never stop. In TOUCH the four pads take hold of them — the upper pad's X is the head position and its Y the speed, the lower pad's X the width of the loop (0.1 to 20 s); the left two are the first reel, the right two the second.

Material gets in by **LOAD** (a file) or **REC** (recording). What is recorded is chosen in the **IN** row: GRAIN, the grain output, or the other reel. Recording is a switch: pressing makes a fresh 8 second loop and writes into it until pressed again. **CLEAR** empties the contents; the loop keeps turning in silence.

The looper inputs

Five inputs per reel, each choosing which of the ring's outputs to take (OSC, CHA, AUD, LOOP, or nothing). The gates follow the height the incoming line usually reaches and open at a proportion of it.

SKIP	Writes a bookmark on the rising edge and returns to it on the falling edge — time stretches. No ceiling on the speed.
FLIP	The direction turns over on every rising edge.
VCA	How far up or down from the knob. It changes how the line returned to the ring crosses the threshold, so it changes the timbre, not only the volume.
RATE / SPD	The playback speed, four octaves of travel.
REC	Overwrites the grain output onto the place being read right now — the layers change over, nothing is erased.

When a looper oscillates

Once the gate is fast enough to be a pitch, the repeated fragment becomes one cycle and the reel turns into an oscillator: the fragment is the waveform and the gate speed is the pitch. Patch a reel's own output into its own SKIP to get there.

Length is resolution

The reel has a fixed number of cells; the speed is how far it steps each sample. Eight seconds at normal speed, sixteen at half, thirty-two at a quarter — the slower, the more goes in and the coarser it gets. Change the speed afterwards and pitch and resolution move together, like a tape.

The rest

The filter at the exit

The reels and the petals patched to EX pass through here; what comes out is added by the EX fader and handed back to the ring as the source EX. Past about three quarters of RES the filter stops needing an input and sings by itself.

DUB	A pair of very sharp band passes set 2.51 apart, each clipping on its own — it beats and sings.
HET	A switched capacitor filter whose own clock folds high sounds back down as difference tones. Turning CUT swims them like tuning an old radio.
ANALOG / DIGITAL	ANALOG clips each pole on its own, lopsided, with a slow drift; DIGITAL is flatter and cleaner.

If it looks like too much

Hold MIC, say something, let go, and move the pads. If the ring seizes up, RESET is in the ring, and the switch on the LFO card unplugs the whole of it in one press.

Credits

The ring of five cells and the gate-driven loopers are influenced by instruments designed by Peter Blasser. Nothing of his is reproduced — the models were written from published descriptions of how the circuits behave and tuned by ear. Frīgedæg is independent software, not affiliated with or endorsed by him. The granular vocabulary — AOA and pulsar synthesis — comes from Dennis Gabor, Iannis Xenakis and Curtis Roads.

Written in Swift with AVAudioEngine, no third party audio code. Frīgedæg is Old English for Friday. © k.odk. Sound you make with it is yours.

